

25F-CY-102
Student ID

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Student Name

Lab 14: Open-Ended Lab-II

Objective:

Design a Program Using Arrays and Functions to Solve a Problem of Your Choice.

Equipment Needed:

- Computer with a C++ compiler (e.g., g++, Visual Studio)
- Text editor or Integrated Development Environment (IDE)

Lab Task (Open-Ended):

Create a C++ program that uses an array and at least two functions to solve any problem you choose.

The program must take input from the user and produce meaningful output.

Requirements (Mandatory):

- An array (1D)
- At least two user-defined functions
- A loop (for or while)
- A conditional statement
- Meaningful comments
- User input & output
- Proper program structure (main + functions)

Examples of Problems You Can Choose (Students pick one):

1. Student Marks Analyzer

- Input marks of N students
- Calculate average
- Find highest & lowest
- Display results

2. Array Statistics Tool

- Input numbers into an array

- Compute sum, mean, max, min
- 3. Frequency Counter**
- Count how many times each number appears
- 4. Sorting Program (Bubble/Selection)**
- Sort array
- Display before & after
- 5. Temperature Record System**
- Input daily temperatures
- Display average
- Show days above average
- 6. Simple Inventory System**
- Store product prices
- Calculate total value
- Find most expensive product
- 7. Custom Problem**
- Students may design their own creative problem involving arrays and functions.

Submission Requirements

- Source code file (lab1.cpp)
- Screenshot of program output
- Short report (1 page):
 - o Problem description
 - o Logic flow
 - o Screenshot
 - o Conclusion